Differentiation: The Quotient Rule

Questions 1–34

Question 1. Standard Functions Find $f'(x)$ and simplify:

a) $f(x) = \frac{3x+1}{2x-1}$
b) $f(x) = \frac{x}{1-4x}$
c) $f(x) = \frac{3x-3}{3x-1}$
d) $f(x) = \frac{e^x}{x-3}$
e) $f(x) = \frac{\ln x}{x^3}$
f) $f(x) = \frac{\sin(10x)}{2x^3}$
g) $f(x) = \frac{5^x}{x^2}$
h) $f(x) = \frac{e^{2x}}{\cos x}$
i) $f(x) = \frac{\ln 5x}{2x+1}$
j) $f(x) = \frac{x^2-1}{\sin x}$

Question 2. Given that $y = \frac{x-5}{\sqrt{x}}$ where $x \neq 0$, show that

$$\frac{dy}{dx} = \frac{x+p}{qx\sqrt{x}}$$

where p and q are integers to be found.

Question 3. Given that $y = \frac{3^x}{x^2 \ln 3}$ where $x \neq 0$, show that

$$\frac{dy}{dx} = \frac{3^x(x \ln 3 - k)}{x^3 \ln 3}$$

where k is an integer to be found.

Question 4. Simplify each of the following:

a) $\frac{(x+1)^2-3x(x+1)}{(x+1)^4}$

b) $\frac{(x-4)^2-2(x+1)(x-4)}{(x-4)^4}$

c) $\frac{2(x+2)^3-4(7x+1)(x+2)^2}{(x+2)^6}$

d) $\frac{4(x+3)^3(5x-1)^3-3(x+3)^4(5x-1)^2}{(5x-1)^6}$

e) $\frac{-6(1-2x)^3(2x+5)^3+6(1-2x)^4(2x+5)^2}{(1-2x)^8}$

Question 5. Show that

$$\frac{(x+1)^{\frac{1}{2}}e^x - \frac{1}{2}(x+1)^{-\frac{1}{2}}e^x}{x+1}$$

may be rearranged to

$$\frac{e^x(1+2x)}{2(x+1)^{\frac{3}{2}}}$$

Question 6. Find $f'(x)$ and simplify:

a) $f(x) = \frac{4x}{(x+1)^2}$

b) $f(x) = \frac{9x-2}{(1+2x)^3}$

c) $f(x) = \frac{e^{4x}}{(x+1)^2}$

Question 7. Find the x-value of the stationary points of the following:

a) $y = \frac{4x-35}{(2x+5)^4}$

b) $y = \frac{(3-2x)^2}{(1+x)^3}$

c) $y = \frac{e^{\frac{1}{2}x}}{2(x-2)^2}$

d) $y = \frac{e^{ax}}{(x+a)^2}, \quad a \neq 0$

Question 8. Reciprocal Trigonometric Functions By writing each first as a quotient, show that:

a) $\frac{d}{dx}(\tan x) = \sec^2 x$

b) $\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$

c) $\frac{d}{dx}(\sec x) = \sec x \tan x$

Question 9. Find $\frac{dy}{dx}$ for each of the following:

a) $y = \tan 3x$
b) $y = \operatorname{cosec} 2x$
c) $y = \sec \frac{1}{3}x$
d) $y = -2\tan \pi x$
e) $y = \tan(-\frac{1}{5}x)$
f) $y = -\sec 9x$
g) $y = \operatorname{cosec}(-\frac{1}{2}x)$

Question 10. Mix of Functions Find when the curve $y = \frac{2x-1}{4x^2+3}$ is decreasing, giving your answer in interval notation.

Question 11. Find any stationary points of the curve $y = \frac{2x-4}{(4x-7)^2}$ and determine their nature.

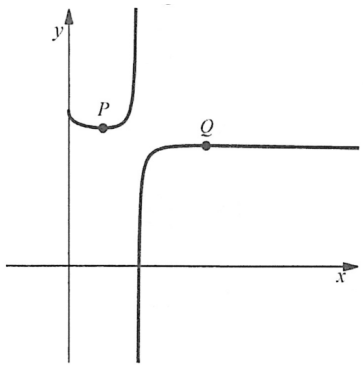
Question 12. $f(x) = -\frac{1}{2}\sec \pi x$, $0 < x < 2$.

a) Find the stationary point on the curve $y = f(x)$.

b) Determine its type.

Question 13. Prerequisite knowledge for Q13: Points of Inflection Show that the curve $y = \frac{\ln x}{x^2}$, $x > 0$ has exactly one single non-stationary point of inflection.

Question 14. The curve $y = \frac{2\sqrt{x}+3x-9}{x-2}$, $x > 0$ is shown on the diagram below.



The stationary points are labelled P and Q.

a) Find the coordinates of P and Q.

b) Find the equation of the straight line passing through P and Q.

A triangle is bounded by the line passing through the points P and Q, the x-axis and the y-axis.

c) Find the area of the triangle.

Question 15. Given $f(x) = \frac{ax-1}{x^2-a}$ show that the gradient at $x = 0$ is always -1.

Question 16. A function is defined by $f(x) = \frac{3}{k+x^2}$, $x > 0$ and k is a positive constant.

a) Show that f is a decreasing function for all values of k.

The point of inflection on the curve is at $x = \frac{2}{\sqrt{3}}$.

b) Find the value of k.

Question 17. Given that $f(x) = \frac{e^{-x}}{k+10x^2}$ where k is a positive constant,

a) find $f'(x)$.

b) Hence find the range of values of k such that $y = f(x)$ has at least one stationary point.

Question 18. The Chain Rule Find $\frac{dy}{dx}$ for each of the following:

a) $y = \tan(x^2 - 1)$

b) $y = \text{cosec}(e^{2x})$

c) $y = \sec(\sin x)$

Question 19. Show that the curve $y = \frac{e^{3x-7}}{(2x-3)^2}$ has exactly one stationary point.

Question 20. $f(x) = \frac{(x^2+1)^2}{10(1-2x)}$, $x \neq k$.

a) State the value of k.
b) Show that the x-values of the maximum and minimum point on the curve $y = f(x)$ satisfy the equation $3x^2 - 2x - 1 = 0$.
c) Hence find the exact values of the maximum and minimum points.
d) Sketch $f(x)$.

Question 21. Find $\frac{dy}{dx}$ for each of the following:

a) $y = \frac{e^{x^2}}{x^2-1}$

b) $y = \frac{\sin(x^3)}{e^{2x+1}}$

c) $y = \frac{3^{\cos x}}{x^2}$

Question 22. It is given that $f(x) = (\frac{x^2-1}{x-3})^4$, where $x \neq 3$.

a) Find $\frac{d}{dx}(f(x))$.

b) Hence show that the tangent line to the curve $y = f(x)$ at the point $x = 2$ is $y - 756x + 1431 = 0$.

Question 23. It is given that $f(x) = (\frac{\ln x}{x^2})^2$, where $x > 0$.

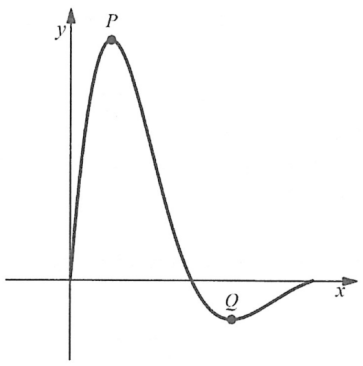
a) Find $\frac{d}{dx}(f(x))$.

b) Hence show that the curve $y = f(x)$ has a stationary point at $x = 1$.

Question 24. The sketch below shows the curve

$$y = \frac{5\sin 3x}{e^{\sqrt{3}x+2}}$$

for $0 \leq x \leq \frac{2\pi}{3}$.



The stationary points are labelled P and Q.

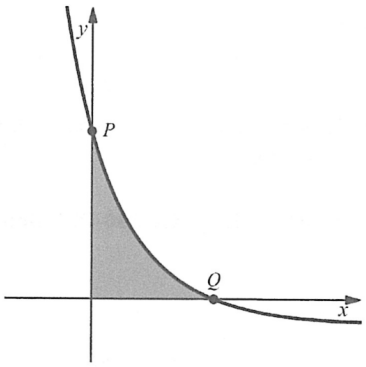
a) Show that the x coordinates of the stationary points are solutions to the equation $\tan 3x = \sqrt{3}$.

b) Find the x values of the stationary points of the curve.

Question 25. $f(x) = \frac{\ln(x+1)}{(x+1)^2}$, $x \geq 0$.

a) Show that $f'(x) = \frac{1-2\ln(x+1)}{(x+1)^3}$.

The diagram below shows the area enclosed by the curve $y = \frac{1-2\ln(x+1)}{(x+1)^3}$ and the x-axis and y-axis.



b) Show that the shaded region is $\frac{1}{2e}$.

Question 26. Prerequisite knowledge for Q26–Q29: The Product Rule Two functions

$$f(x) = (1-2x)(2x^2+ax+b) \qquad \text{and} \qquad g(x) = \frac{9(ax+b)}{x^2+2}$$

where a and b are integers. Given that $f'(1) = g'(1) = -8$ find the values of a and b.

Question 27. Find $f'(x)$ for each of the following:

a) $f(x) = \frac{x^2 e^x}{2x-1}$

b) $f(x) = \frac{2^x \sin x}{\sqrt{x}}$

c) $f(x) = \frac{x \ln x}{(x-1)^2}$

Question 28. A curve has equation $y = \frac{(x^2-2x-3)e^x}{\sqrt{2x+1}}$. Show that the equation of the tangent to the curve at the point $x = 0$ is $y = -2x - 3$.

Question 29. A curve is given by $y = \frac{e^{\frac{1}{2}x}}{x^2+2x+1}$ for $x \geq 0$.

a) Find the x-value of the single stationary point on the curve.

b) Show that $\frac{d^2y}{dx^2} = \frac{e^{\frac{1}{2}x}(x^2-6x+17)}{4(x+1)^4}$.

c) Hence show why the curve has no points of inflection.

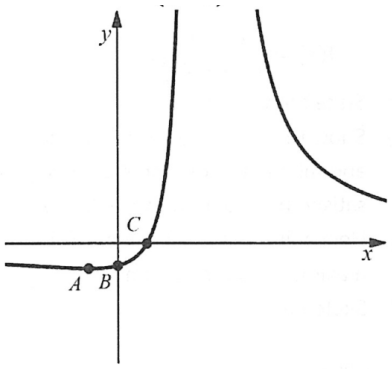
Question 30. Prerequisite knowledge for Q30–Q31: Double Angle Formulae Show that $\frac{d}{dx}\left(\frac{\sin 2x}{\cos x}\right) = 2\cos x$.

Question 31. Show that

$$\frac{d}{dx}\left(\frac{2\sin x}{5\sin x + 5\cos x}\right) = \frac{p}{q(1+\sin 2x)}$$

where p and q are integers and $\frac{p}{q}$ is in its most simplified form.

Question 32. Prerequisite knowledge for Q32–Q34: Domain and Range of Functions The sketch below shows the curve $y = f(x)$ where $f(x) = \frac{x-1}{(x-3)^2}$, $x \neq k$.



The stationary point is labelled A. The intercepts with the axes are labelled B and Q.

a) State the value of k.
b) Find the coordinates of the points B and C.
c) Find the coordinates of A.
d) Sketch on separate axes, labelling the new values of A, B and C:
   i) $f(x) + 1$
   ii) $f(2x)$

Question 33. Given that $f(x) = \frac{2\ln x - 5}{\ln x - 1}$.

a) State the maximum domain of f.

b) Show that $f(x)$ is an increasing function.

c) Find the set of values for which $f(x) < 0$.

Question 34. Given $f(x) = \frac{2^x}{x-1}$.

a) State the maximum possible domain of f.

b) Find $f'(x)$ and hence find the coordinate of the single stationary point of f giving your answer to 3 significant figures.

c) Hence find the range of f.